Parnassus: Fast Detector Simulation and Reconstruction in Pure Python

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Abstract

We present the public software release of Parnassus, a pure-Python, GPU-compatible framework for fast detector simulation and reconstruction in High Energy Physics. Parnassus replaces the computationally expensive Geant4 simulation and reconstruction chain with two interchangeable generator backends: a **neural backend** based on conditional flow matching models that map truth-level particles directly to detector-level particle-flow objects, and a **parametric backend** (TorchDelphes) that re-implements the Delphes fast simulation chain in pure PyTorch. Both backends share the same process-agnostic and detector-agnostic API: users select a detector card—analogous to choosing a detector card in Delphes—and the same tool can be applied to arbitrary physics processes without retraining. Unlike C++/ROOT-based tools such as Delphes, Parnassus runs natively on GPU (CUDA and Apple MPS) and requires no ROOT installation. The current release ships with CMS and ATLAS parametric cards validated against C++ Delphes 3.5.0, and a CMS neural model trained on 2011 open data; additional detector models will be provided in future releases. We describe the installation, command-line and Python API, configuration system, and demonstrate the framework on Standard Model and BSM processes.

Keywords: fast simulation, detector simulation, particle flow, generative models, flow matching, diffusion models, high energy physics

PROGRAM SUMMARY

Program title:	Parnassus
CPC Library link to program files:	(to be added by Technical Editor)
Developer's repository link:	https://github.com/parnassus-hep/parnassus
Licensing provisions:	MIT
Programming language:	Python ($\geq$ 3.12)
Nature of problem:	Full detector simulation and reconstruction based on Geant4 is computationally expensive, often dominating the computing budget of large-scale High Energy Physics analyses. A fast, accurate surrogate is needed that maps truth-level (generator-level) particles directly to detector-level particle-flow objects, preserving the fidelity of the full chain while being orders of magnitude faster.
Solution method:	Parnassus offers two interchangeable generator backends sharing a common API. The *neural backend* trains conditional flow matching models [7, 8] on full simulation and reconstruction data; the model is conditioned on truth-level particle properties and generates detector-level particle-flow

objects including kinematics, classification, and track impact parameters. The *parametric backend*, TorchDelphes, re-implements the Delphes [25] fast-simulation chain (particle propagation, tracking efficiency, momentum smearing, calorimeter response, energy-flow merging) in pure PyTorch, validated against C++ Delphes 3.5.0 for CMS and ATLAS cards. Both backends are followed by physics-based post-processing (jet clustering via FastJet [17], lepton isolation). Users select a detector card—analogous to a Delphes card—to target a specific experiment.

Additional comments including restrictions and unusual features:
The framework is implemented entirely in Python with PyTorch and runs natively on CPU, CUDA GPU, and Apple MPS. It has no dependency on ROOT; output to ROOT format is optionally handled via `uproot` [24]. The current release ships with CMS and ATLAS parametric (TorchDelphes) cards and a CMS neural model trained on 2011 open data [6]; additional detector models for other experiments will be added in future releases. The neural CMS model supports up to 400 truth particles per event; events exceeding this limit retain the 400 highest-p_{T} particles.

1 Introduction

Parnassus is a **pure-Python**, **fully GPU-compatible** framework for **fast detector simulation and reconstruction** in High Energy Physics (HEP). It replaces the computationally expensive full Geant4 [3] simulation and reconstruction chain with generative models that map truth-level (generator-level) particles directly to detector-level particle-flow [4, 5] (pflow) objects, bypassing intermediate steps such as hit simulation, digitisation, and track reconstruction. The framework is both **process-agnostic** and **detector-agnostic**: users select a detector card to target a specific experiment, and the same tool can be applied to arbitrary physics processes by simply providing different input truth events.

Parnassus exposes two interchangeable **generator backends** sharing a common API:

- A **neural backend** based on conditional flow matching [7, 8] trained on full simulation and reconstruction data, providing high-fidelity learned detector response.
- A **parametric backend**, TorchDelphes, which re-implements the Delphes [25] fast-simulation chain (particle propagation, tracking efficiency, momentum smearing, calorimeter response, energy-flow merging) in pure PyTorch. The implementation is validated against C++ Delphes 3.5.0 for both CMS and ATLAS detector cards and provides a GPU-native drop-in alternative to running C++ Delphes.

Unlike traditional fast simulation tools such as Delphes [25], which are written in C++, require the ROOT framework [19], and run only on CPU, Parnassus is implemented entirely in Python with PyTorch [22] and runs natively on GPU (CUDA and Apple MPS), enabling seamless integration into modern Python-based analysis workflows and significant acceleration on GPU hardware. ROOT I/O is supported via `uproot` [24] for writing output files, but is entirely optional — Parnassus has no dependency on ROOT itself.

Users select a **detector card** — analogous to choosing a detector card in Delphes — to target a specific experiment. The current release ships with CMS and ATLAS TorchDelphes parametric cards, and a CMS neural model trained on 2011 open data [6]; additional detector models will be added in future releases.

The physics methodology underlying Parnassus has been described in detail in Refs. [1, 2], where we demonstrated that the approach reproduces full CMS simulation and reconstruction across a wide range of Standard Model and BSM processes. In this document we focus on the **public software release** — its installation, usage, and API — to enable the broader HEP community to adopt neural fast simulation and reconstruction in their workflows.

The neural backbone uses conditional flow matching [7, 8], a continuous-time generative modelling framework closely related to score-based diffusion models [9, 10], sampled via Euler integration of the learned vector field at inference time. The models are conditioned on truth-level particle properties, making the generation inherently process-agnostic without requiring retraining for new physics scenarios.

1.1 Design goals

Speed Orders of magnitude faster than full simulation and reconstruction by replacing the detector response with a learned generative model. Fully GPU-compatible for additional acceleration.

Pure Python
Implemented entirely in Python with no compiled dependencies beyond PyTorch. Unlike C++/ROOT-based tools such as DELPHES [25], Parnassus integrates naturally into modern Python analysis ecosystems. ROOT output is available via `uproot` but is not required.

Fidelity Each detector model is trained on full simulation and reconstruction data to reproduce realistic detector-level distributions including particle kinematics, vertex positions, impact parameters, and particle identification. Excellent agreement with full simulation has been demonstrated across diverse SM and BSM processes [1, 2].

Flexibility Accept truth-level input from any source — PYTHIA8 [11] on-the-fly generation, HepMC [12] files from external generators (MADGRAPH5_AMC@NLO [13], SHERPA [14], HERWIG [15], POWHEG [16]), or custom formats. Users select a detector model to target a specific experiment, analogous to choosing a detector card in DELPHES.

Completeness
Full post-processing pipeline including jet clustering (FASTJET [17]) and lepton isolation, with optional output to ROOT format via `uproot` [24].

1.2 Output quantities

Given a set of truth-level particles for each event, Parnassus generates the quantities listed in Table 1.

Table 1: Output quantities produced by Parnassus for each event.

Output	Description
Particle-flow particles	Full kinematics (p_T, η, ϕ), vertex (v_x, v_y, v_z), class, PDG ID
Impact parameters	d_0, z_0 and their errors for charged particles
Jets	Clustered with configurable algorithms (anti-k_t [18], gen-k_t); substructure (D_2 [20], C
Lepton isolation	Isolation scores and p_T sums in configurable ΔR cones
Event-level quantities	H_T, $\vec{E}_\mathrm{T}^\mathrm{miss}$ components, particle multiplicity

2 Architecture overview

Parnassus passes truth-level events through a detector model followed by physics-based post-processing. The detector model can be either a neural backend or the TORCHDELPHES paramet-

ric backend; both share the same input and output interface so that downstream post-processing is unchanged. The pipeline is illustrated in Fig. 1.

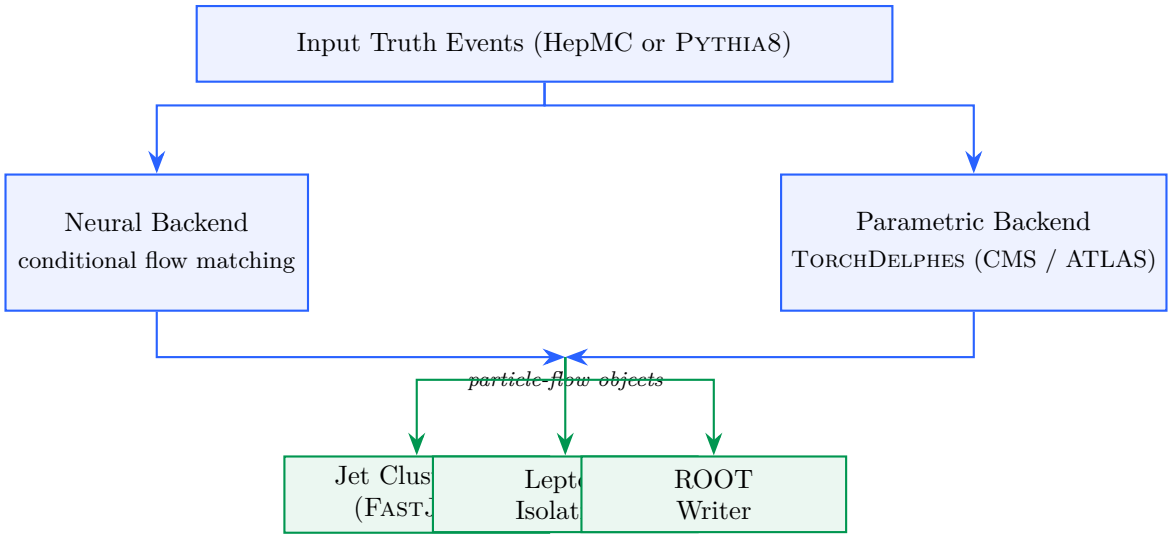

Figure 1: Parnassus generation pipeline. Truth-level events are routed through either the neural backend (conditional flow matching) or the parametric backend (TorchDelphes), both producing particle-flow objects. Physics-based post-processing (green) computes derived quantities and writes the final output.

The **neural backend** is a continuous-time generative model trained with conditional flow matching [7, 8] and sampled via Euler integration of the learned vector field. The model is conditioned on truth-level particle information, making the generation process inherently dependent on the input physics process without being tied to any specific one. The number of integration steps is configurable at runtime via the `--num_steps` (`-n`) flag, allowing users to trade generation speed for quality.

The **parametric backend**, TorchDelphes, is a pure-PyTorch re-implementation of the Delphes [25] fast-simulation chain. It models particle propagation through the magnetic field, tracking efficiency, momentum smearing, calorimeter response, and energy-flow merging, with per-module parameters loaded from the selected detector card. The current release provides CMS and ATLAS cards ported from their respective Delphes TCL cards and validated against C++ Delphes 3.5.0. Being implemented in pure PyTorch, TorchDelphes runs natively on GPU and shares the same input and output conventions as the neural backend.

Both backends expose the same output contract (particle-flow kinematics, vertex positions, particle classification, and, where supported, track impact parameters), so downstream post-processing is identical.

3 Installation and quick start

3.1 Installation

```
# Clone the repository
git clone https://github.com/parnassus-hep/parnassus.git
cd parnassus

# Install with pip (requires Python 3.12)
pip install .

# For development
```

```
pip install -e ".[dev]"
```

Listing 1: Installation commands.

3.2 Requirements

- Python $\geq 3.12, < 3.13$
- PyTorch [22] $\geq 2.6.0$
- Pythia8mc [11] 8.316.0
- PyHepMC [12] $\geq 2.14.0$
- FastJet [17] $\geq 3.4.3.1$
- EnergyFlow [23] $\geq 1.4.0$
- Uproot [24] $\geq 5.5.2$ *(optional, for ROOT output)*

3.3 Initialise a configuration file

```
parnassus init .
```

This copies the default `default_config.yaml` to the current directory. The default configuration uses the `cms_2011_flow_v00` neural detector model with anti-k_t jet clustering and lepton isolation. To target a different detector or to switch to the parametric (TorchDelphes) backend, change the `generator.type` and `generator.name` fields in the YAML configuration. The two generator backends are configured as shown in Listing 2.

```
# --- Neural backend (conditional flow matching) ---
generator:
  type: "neural"
  name: "cms_2011_flow_v00"  # trained CMS model
  num_steps: 50
  batch_size: 2000
  device: "cpu"              # "cpu", "cuda", or "mps"

# --- Parametric backend (TorchDelphes) ---
generator:
  type: "parametric"
  name: "cms"                # "cms" or "atlas" detector card
  seed: 42                   # optional, for reproducibility
```

Listing 2: Switching between the neural and parametric (TorchDelphes) generator backends.

3.4 Run with a HepMC input file

```
parnassus run \
  -c default_config.yaml \
  -i events.hepmc \
  -ne 1000 \
  -bs 2000 \
  -o output.root
```

Listing 3: Basic command-line invocation.

The available command-line flags are listed in Table 2.

Table 2: Command-line flags for `parnassus run`.

Flag	Long form	Description
`-c`	`--config`	Path to YAML configuration file
`-i`	`--input_path`	Input HepMC or Pythia `.cmnd` file
`-o`	`--output_path`	Output ROOT file path
`-ne`	`--num_events`	Number of events to process
`-bs`	`--batch_size`	Batch size for neural generation
`-n`	`--num_steps`	Neural-backend Euler integration steps (default: 50)

3.5 Run with Pythia8 on-the-fly generation

To generate truth-level events on-the-fly with PYTHIA8, create a `.cmnd` steering card and pass it as the input file. Parnassus detects `.cmnd` files automatically:

```
parnassus run \
  -c default_config.yaml \
  -i ttbar.cmnd \
  -ne 5000 \
  -bs 2000 \
  -o ttbar_fastsim.root
```

4 Demonstration: SM and BSM processes

The neural backend is **process-agnostic** — it learns the detector response as a function of truth-level particle properties, not the physics process itself — and the parametric (TORCHDELPHES) backend is process-agnostic by construction. We have previously demonstrated [1, 2] that Parnassus reproduces full simulation and reconstruction across a wide range of SM and BSM processes. Here we show two representative examples to illustrate the API in action: a Standard Model process ($H \to ZZ \to 4\ell$) and a BSM exotic Higgs decay ($H \to aa \to gg\mu\mu$). We then re-run the SM example with the parametric TORCHDELPHES backend to demonstrate that the two generator models can be swapped without touching the rest of the pipeline. The kinematic and class-fraction summaries that follow (Sections 4.4 onwards) are produced with the neural backend.

The current CMS neural model supports up to 400 truth particles per event. For events exceeding this limit, Parnassus retains the 400 highest-p_{T} particles so that no events are discarded.

4.1 Standard Model: $H \to ZZ \to 4\ell$

We process 100 events from a HepMC input file produced by PYTHIA8 [11]:

```
parnassus run \
  -c default_config.yaml \
  -i h4l_events.hepmc \
  -ne 100 \
  -o h4l_output.root
```

4.2 BSM: $H \to aa \to gg\mu\mu$

For the BSM example, we generate exotic Higgs decays on-the-fly with Pythia8. The Higgs decays to a pair of light pseudoscalars a ($m_a = 20\,\text{GeV}$), each decaying to either gg or $\mu^+\mu^-$, producing a mixed final state of jets and muon pairs:

```
! haa_ggmumu.cmnd -- BSM exotic Higgs decay
Beams:eCM = 13000.
Beams:idA = 2212
Beams:idB = 2212

! BSM Higgs production via gluon fusion
Higgs:useBSM = on
HiggsBSM:gg2H2 = on
35:m0 = 125.0                          ! Higgs mass

! SM-like couplings for production
HiggsH2:coup2u = 1.0
HiggsH2:coup2d = 1.0
HiggsH2:coup2A3A3 = 1.0                ! H -> aa coupling

! Force H -> a a only
35:onMode = off
35:onIfAll = 36 36

! Light pseudoscalar a properties (PDG ID 36)
36:m0 = 20.0                           ! mass = 20 GeV
HiggsA3:coup2d = 1.0
HiggsA3:coup2u = 1.0
HiggsA3:coup2l = 1.0

! Force a -> gg or mumu only
36:onMode = off
36:onIfAny = 21 13
```

Listing 4: Pythia8 steering card for BSM $H \to aa \to gg\mu\mu$.

```
parnassus run \
  -c default_config.yaml \
  -i haa_ggmumu.cmnd \
  -ne 1000 \
  -o haa_ggmumu_output.root
```

This demonstrates running Parnassus on a BSM signal process not present in the training data. The mixed $gg\mu\mu$ final state probes the model's ability to simultaneously generate hadronic jets and isolated muons within the same event.

4.3 Swapping backends: same events, parametric TorchDelphes

The two generator backends are swapped purely through the configuration file: neither the input HepMC file, the post-processing pipelines, nor the output schema change. Listing 5 shows a `cms_parametric.yaml` that re-runs the same $H \to ZZ \to 4\ell$ events through the parametric (TorchDelphes) backend with the CMS card.

```
# cms_parametric.yaml
dataset:
  file_path: ""
```

```
  num_events: 100

# Swap only this block: neural -> parametric
generator:
  type: "parametric"    # was: "neural"
  name: "cms"           # was: "cms_2011_flow_v00"
  seed: 42

pipelines:
  TruthJetsAntiKt05: { type: cluster, collection: truth, dr: 0.5,
                       algorithm: antikt, pt_min: 10, nconst_min: 2 }
  PflowJetsAntiKt05: { type: cluster, collection: pflow, dr: 0.5,
                       algorithm: antikt, pt_min: 10, nconst_min: 2 }
  ElectronIsolation: { type: isolation, collection: electrons, dr: 0.4
      }
  MuonIsolation:     { type: isolation, collection: muons,     dr: 0.4
      }

output:
  file_path: ""
  format: default
```

Listing 5: Configuration for running TorchDelphes with the CMS card on the same $H \to 4\ell$ events.

```
parnassus run \
  -c cms_parametric.yaml \
  -i h4l_events.hepmc \
  -ne 100 \
  -o h4l_torchdelphes.root
```

Listing 6: Running the same events through the parametric backend.

The two output ROOT files (`h4l_output.root` from the neural backend and `h4l_torchdelphes.root` from TorchDelphes) follow the same schema, so the same downstream analysis code reads either of them unchanged. Switching to the ATLAS card requires only changing `generator.name` from `"cms"` to `"atlas"` in the same configuration file — analogous to selecting a different detector card in Delphes.

4.4 Kinematic distributions

Fig. 2 compares the particle-flow kinematic distributions for both processes. The p_{T} spectra follow the expected steeply falling shape, the η distributions reflect the detector acceptance, and the multiplicity distributions reflect the different underlying event structures. The BSM $H \to aa$ process shows higher multiplicity due to the hadronic activity from the $a \to gg$ decays.

4.5 Event display

Fig. 3 shows a representative $H \to 4\ell$ event in the η–ϕ plane, illustrating how Parnassus transforms truth-level particles into detector-level pflow objects.

4.6 Aggregate statistics

Table 3 compares the aggregate statistics for both processes. The BSM process shows higher truth-particle multiplicity (163 ± 70) due to the $a \to gg$ hadronic decays, while the pflow output multiplicity is comparable across both processes.

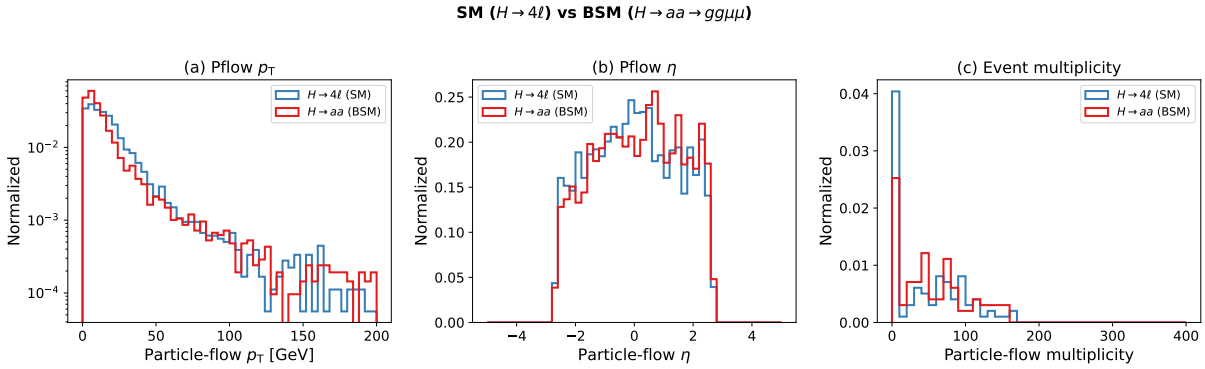

Figure 2: Particle-flow kinematic distributions comparing SM ($H\to4\ell$, 99 events) and BSM ($H\to aa\to gg\mu\mu$, 99 events) processes. (a) p_{T} in log scale, (b) pseudorapidity η, (c) particle-flow multiplicity per event. All distributions are normalised to unit area.

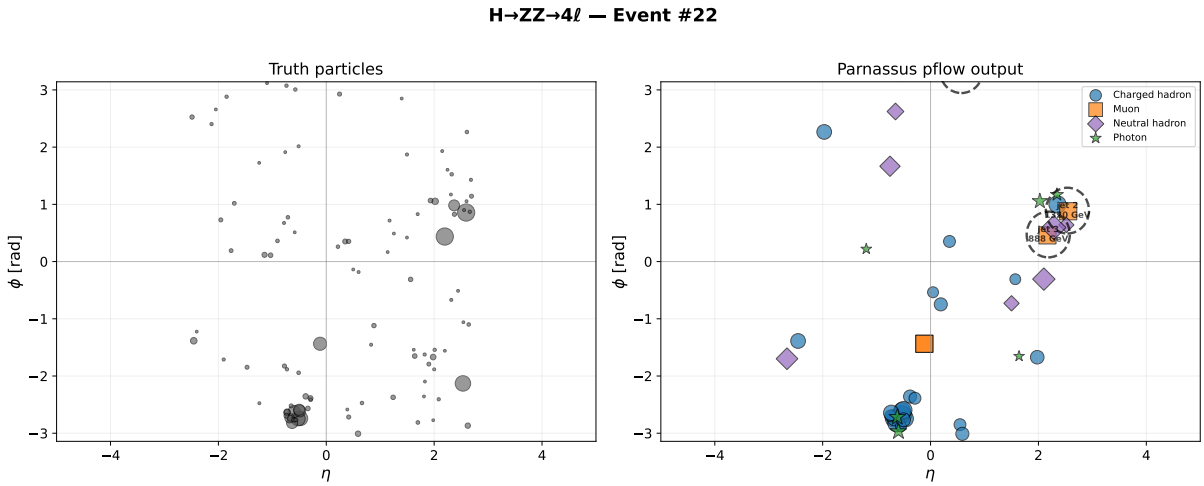

Figure 3: Event display for $H\to ZZ\to4\ell$: truth particles (left) and Parnassus pflow output (right) in the η–ϕ plane. Marker size scales with p_{T}; dashed circles indicate anti-k_t jets ($R=0.5$). Particle classes: charged hadrons (blue circles), electrons (red triangles), muons (orange squares), neutral hadrons (purple diamonds), photons (green stars).

Table 3: Aggregate statistics for SM and BSM processes (mean ± std).

Quantity	$H\to4\ell$ **(99 evt)**	$H\to aa$ **(99 evt)**
Truth particles/event	97.2 ± 46.0	163.4 ± 69.9
Pflow particles/event	46.3 ± 47.2	53.6 ± 46.4

4.7 Particle class distribution

Fig. 4 compares the particle class composition between the two processes. Both show similar class fractions dominated by charged hadrons (∼60%), consistent with the process-agnostic nature of the model.

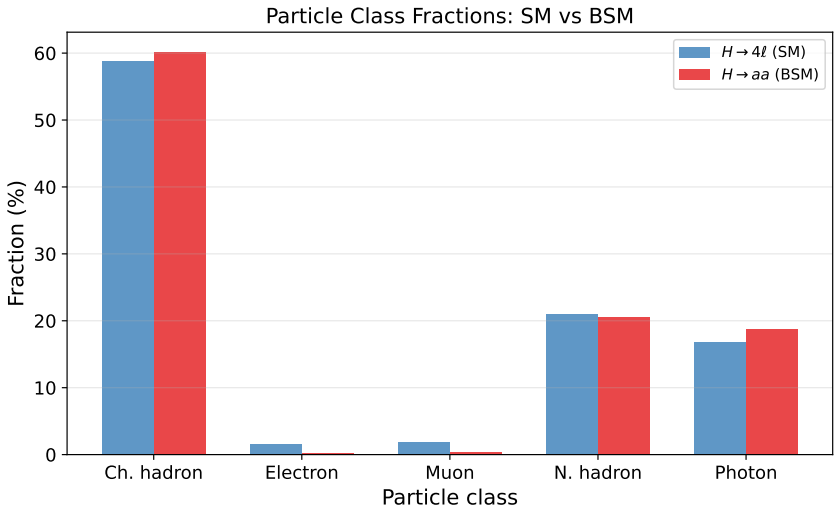

Figure 4: Particle class fractions for SM ($H \to 4\ell$) and BSM ($H \to aa \to gg\mu\mu$) processes. The model produces consistent class compositions regardless of the input physics process.

5 Neural backend: integration-step convergence

This section applies only to the neural backend. The number of Euler integration steps N used to sample the conditional flow-matching model controls the trade-off between generation quality and speed. Fig. 5 shows the convergence of particle-flow kinematic distributions as a function of N for the $H \to 4\ell$ sample.

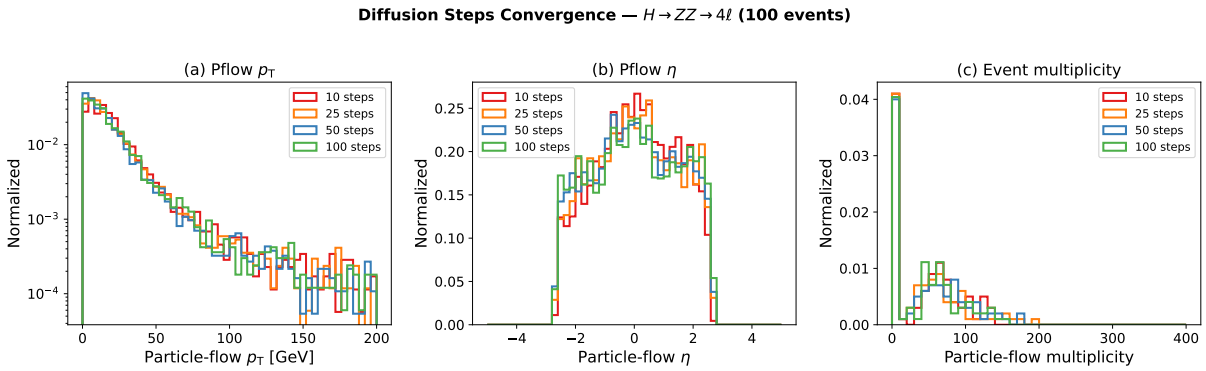

Figure 5: Integration-step convergence of the neural backend for $H \to ZZ \to 4\ell$ (100 events). (a) Pflow p_T, (b) pflow η, (c) event multiplicity. Distributions are stable from $N = 25$ onward.

Fig. 6 shows that particle class fractions are also stable across step counts.

Table 4 summarises the generation time as a function of the number of integration steps on CPU.

The default of 50 steps provides a good balance of quality and speed. For quick validation runs, 25 steps yields nearly identical distributions at lower cost.

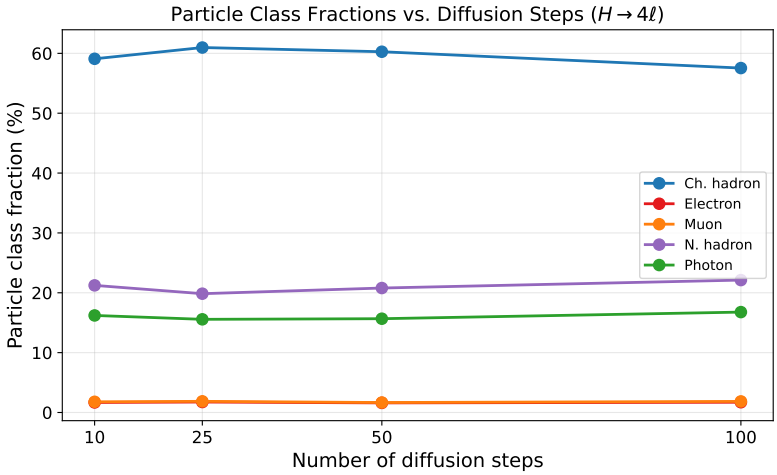

Figure 6: Particle class fractions vs. number of Euler integration steps for the neural backend. All classes converge by $N = 25$.

Table 4: Neural-backend generation time vs. number of Euler integration steps (100 events, CPU).

Steps N	Wall time
10	$\sim$90 s
25	$\sim$90 s
50	$\sim$130 s
100	$\sim$320 s

6 Python API

For programmatic usage, Parnassus provides a clean Python API:

```python
from pathlib import Path
from parnassus.configs import Config
from parnassus.pipelines import (
    GenerationPipeline,
    JetClusteringPipeline,
    IsolationPipeline,
)
from parnassus.configs.pipeline import (
    JetClusteringConfig,
    IsolationConfig,
)
from parnassus.writers import RootWriter

# Load configuration
config = Config.from_yaml("my_config.yaml")

# Override settings programmatically
config.dataset_config.file_path = (
    Path("my_events.hepmc").absolute()
)
config.dataset_config.num_events = 5000

# Run generation
```

```python
pipeline = GenerationPipeline(config)
gen_events, accessors_dict = pipeline.run()

# Post-processing
for pipeline_config in config.pipeline_configs:
    if isinstance(pipeline_config, JetClusteringConfig):
        jet_pipeline = JetClusteringPipeline(pipeline_config)
        jet_pipeline.process(gen_events)
    elif isinstance(pipeline_config, IsolationConfig):
        iso_pipeline = IsolationPipeline(pipeline_config)
        iso_pipeline.process(gen_events)

# Inspect generated data
for event in gen_events[:5]:
    print(f"Event {event.event_number}:")
    n_truth = len(event.truth_particles.pt)
    n_pflow = len(event.pflow_particles.pt)
    print(f"  Truth particles: {n_truth}")
    print(f"  Pflow particles: {n_pflow}")
    print(f"  HT = {event.ht:.1f} GeV")

# Write to ROOT
writer = RootWriter(config.writer_config)
writer.write(gen_events)
```

Listing 7: Programmatic usage of the generation pipeline.

6.1 Reading output files

```python
import uproot
import numpy as np

f = uproot.open("output.root")
tree = f["Parnassus"]

# Access pflow jet pT
jet_pt = tree["PflowJetsAntiKt05.pt"].array()

# Access electron isolation
e_iso = tree["Electrons.iso_var"].array()

# Access impact parameters
d0 = tree["Pflow.d0"].array()
z0 = tree["Pflow.z0"].array()
```

Listing 8: Reading the output ROOT file with `uproot`.

7 Configuration reference

```yaml
# -- Dataset ------------------------------------------
dataset:
  file_path: ""               # Path to .hepmc or .cmnd file
  num_events: 1000            # Number of events to process
  entry_start: 0              # Starting event index (HepMC)
```

```yaml

# -- Generator (choose one backend) --------------------
# Option A: Neural backend (conditional flow matching)
generator:
  type: "neural"              # Select the neural backend
  name: "cms_2011_flow_v00"   # Detector model (like a Delphes card)
  num_steps: 50               # Number of flow-matching integration
      steps
  batch_size: 2000            # Events per batch
  device: "cpu"               # "cpu", "cuda", or "mps"

# Option B: Parametric backend (TorchDelphes)
# generator:
#   type: "parametric"        # Select the TorchDelphes backend
#   name: "cms"               # Detector card: "cms" or "atlas"
#   max_particles: 10000      # Maximum particles per event
#   seed: 42                  # Random seed for stochastic smearing

# -- Post-processing Pipelines -------------------------
pipelines:
  TruthJetsAntiKt05:
    type: "cluster"
    collection: truth
    dr: 0.5
    algorithm: antikt
    pt_min: 10
    nconst_min: 2
  PflowJetsAntiKt05:
    type: "cluster"
    collection: pflow
    dr: 0.5
    algorithm: antikt
    pt_min: 10
    nconst_min: 2
  ElectronIsolation:
    type: "isolation"
    collection: "electrons"
    dr: 0.4
  MuonIsolation:
    type: "isolation"
    collection: "muons"
    dr: 0.4

# -- Output --------------------------------------------
output:
  file_path: ""               # Output ROOT file path
  format: default
```

Listing 9: Annotated default configuration.

8 Conclusion

We have presented the public release of Parnassus, a pure-Python, GPU-compatible framework for fast simulation and reconstruction in High Energy Physics. Unlike traditional C++/ROOT-based tools such as Delphes [25], Parnassus runs entirely in Python with PyTorch, supports GPU acceleration (CUDA and Apple MPS), and requires no ROOT installation — ROOT

output is optionally handled via `uproot`. The software provides a simple command-line and Python API for generating detector-level particle-flow objects from truth-level input, with full post-processing including jet clustering and lepton isolation.

Parnassus offers two interchangeable generator backends that share a common user-facing API: a **neural backend** based on conditional flow matching models trained on full simulation, and a **parametric backend** (TorchDelphes) that re-implements the Delphes fast simulation chain — particle propagation, tracking efficiency, momentum smearing, calorimetry, and energy-flow merging — as pure-PyTorch modules. TorchDelphes has been validated against C++ Delphes 3.5.0 and provides a GPU-native drop-in alternative to C++ Delphes.

The framework is both process-agnostic and detector-agnostic. Users select a detector model — analogous to choosing a detector card in Delphes — to target a specific experiment. The current release ships with CMS and ATLAS parametric cards for the TorchDelphes backend, and a CMS neural model validated against full CMS simulation across diverse SM and BSM processes [1, 2]; additional detector models and neural checkpoints will be provided in future releases.

The source code is publicly available at https://github.com/parnassus-hep/parnassus under the MIT license.

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